

ORIGINAL RESEARCH ARTICLE

Pulsed Field Ablation Versus Sham to Treat Atrial Fibrillation: The PFA-SHAM Randomized Clinical Trial

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BACKGROUND: Catheter ablation for atrial fibrillation (AF) is one of the most common cardiovascular procedures being performed worldwide. Despite the large body of evidence of its effectiveness, with a single exception, prior ablation studies were largely unblinded trials. Accordingly, residual concerns remained about placebo effects, both for AF recurrence and, in particular, on subjective outcomes such as quality of life or anxiety. Here, we compared pulsed field ablation (PFA) with a sham procedure to treat patients with symptomatic AF.

METHODS: This prospective, sham-controlled, single-blind, randomized clinical trial with blinded end-point assessment enrolled patients with AF that was highly symptomatic (Atrial Fibrillation Effect on Quality-of-Life score <50). Patients were assigned 1:1 to PFA or a sham procedure. All participants received implantable cardiac monitors for continuous rhythm monitoring during follow-up. The 6-month co-primary outcomes were (1) time to first recurrence of atrial tachyarrhythmia and (2) changes from baseline in Atrial Fibrillation Effect on Quality-of-Life scores compared between groups. Secondary outcomes were AF burden and psychological distress (assessed by the Hospital Anxiety and Depression Scale [HADS]).

RESULTS: Patients (n=60) were randomized to PFA or sham. At 6 months, the first co-primary end point of AF recurrence was met in 2 patients (6.7%) who underwent PFA and 25 patients (83.3%) who underwent sham (posterior hazard ratio, 19.6 [95% bayesian credible intervals, 6.7–76.9]; posterior probability of superiority >0.99). For the second co-primary end point, Atrial Fibrillation Effect on Quality-of-Life scores showed greater improvement from baseline with PFA than sham (improved by 43.9±18.1 points versus 11.3±27.9 points; posterior median difference, 32.6 [95% bayesian credible interval, 20.2–44.9]; posterior probability of superiority >0.99). AF burden at 6 months was significantly lower in the PFA than the sham group (0 [0–0] versus 0.43 [0.04–3.47]; between group median difference, –0.39 [95% credible interval, –2.5 to –0.1], posterior probability of superiority >0.99). The Hospital Anxiety and Depression Scale score changed by –4 points (–7.8 to –2.0) with PFA and by –0.5 (–4.5 to 1.0) with sham (group median difference, –3.5 [95% credible interval, –6.0 to –1.0]; posterior probability of superiority >0.99).

CONCLUSIONS: In patients with AF, PFA was superior to sham in reducing arrhythmia recurrences and burden and improving quality of life and AF-associated psychological distress.

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Key Words: atrial fibrillation ■ catheter ablation ■ psychological distress ■ quality of life

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Clinical Perspective

What Is New?

- In a randomized trial of patients with symptomatic atrial fibrillation (AF), compared with a sham procedure, pulsed field ablation significantly reduced recurrences of AF, atrial tachycardia, or atrial flutter.
- Pulsed field ablation also produced substantially greater improvements in a disease-specific quality-of-life measure (Atrial Fibrillation Effect on Quality-of-Life score) than the sham procedure, supporting a true treatment effect beyond placebo.
- Pulsed field ablation was also associated with a marked reduction in AF burden and significant improvements in psychological distress, as assessed with the Hospital Anxiety and Depression Scale questionnaire.

What Are the Clinical Implications?

- The study provides the evidence that pulsed-field ablation for AF confers benefits in rhythm control and patient-reported outcomes beyond potential placebo effects.
- Pulsed field ablation represents an effective strategy to reduce arrhythmia burden, improve quality of life, and decrease AF-related anxiety.
- Sham-controlled trial designs may help clarify the true clinical benefits of interventional procedures, particularly for outcomes that can be influenced by symptoms and patient perception.

Among patients with atrial fibrillation (AF), previous randomized controlled trials have shown that catheter ablation is more efficacious than antiarrhythmic drugs in preventing arrhythmia recurrence.^{1–4} This has proved true for both symptomatic paroxysmal and persistent AF, as well as after first-line ablation or second-line ablation after failed drug treatment. However, a mortality benefit from ablation was demonstrated only in patients with AF with heart failure.^{5,6} Accordingly, the primary indication for ablation in AF management is to improve patient symptoms. Although randomized trials have shown that ablation reduces arrhythmia recurrence and improves quality of life and psychological distress, nearly all of these trials were open label, that is, not comparing with a sham procedure.

Because quality of life and psychological distress are subjective parameters susceptible to placebo effects, a criticism that has been levied against such open-label trials is the absence of patient blinding. Furthermore, beyond quality of life, even “hard” end points such as AF recurrences might also be affected by placebo-related mechanisms.⁷ Previous sham-controlled randomized studies of other cardiovascular procedures such as renal denervation or percutaneous coronary intervention in stable coronary artery disease have proved elucidative:

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
AFEQT	Atrial Fibrillation Effect on Quality-of-Life
AFL	atrial flutter
AT	atrial tachycardia
CrI	credible interval
DSMB	Data and Safety Monitoring Board
HADS	Hospital Anxiety and Depression Scale
ICM	implantable cardiac monitor
PFA	pulsed field ablation
PRF-SHAM	Pulsed-Field Ablation Vs. Sham Ablation to Treat AF
PVI	pulmonary vein isolation
REMEDIAL	Randomized Evaluation of the Impact of Catheter Ablation on Psychological Distress in Atrial Fibrillation
SHAM-PVI	A Randomized Sham-Controlled Study of Pulmonary Vein Isolation in Symptomatic Atrial Fibrillation

Sham-controlled trials have effectively distinguished the actual therapeutic impact of the active interventions from placebo effects.^{8–11}

In the only randomized study comparing pulmonary vein isolation (PVI) with a sham procedure, the primary end point was AF recurrence, whereas changes to quality of life were secondary end points for which no formal power calculation was performed and no adjustment for multiplicity was applied.¹² Furthermore, that study used cryotherapy as the energy source for ablation; however, driven largely by procedural workflow and safety considerations, the landscape for AF ablation technologies has evolved to pulsed field ablation (PFA) in many centers. Accordingly, we conducted a randomized controlled trial in patients with highly symptomatic AF in which we compared catheter ablation using a pentaspline PFA catheter with a sham procedure to accurately delineate the impact of PFA on both freedom from AF and symptom relief. Implantable cardiac monitors (ICMs) were implanted in all patients for continuous and accurate rhythm monitoring in follow-up.

METHODS

Study Design

PFA-SHAM (Pulsed-Field Ablation Vs. Sham Ablation to Treat AF) was an investigator-initiated, dual-center, sham-controlled, single-blind, randomized superiority trial with blinded end point (PROBE [Prospective Randomized Open-label Blinded Endpoint]) adjudication. The study design has been described

previously.¹³ The full protocol, including the statistical analysis plan, is provided in the [Appendix in the Supplemental Material](#). Subjects were enrolled in 2 centers in the Czech Republic. The study complied with the Declaration of Helsinki and was approved by the multicenter Ethics Committee of the University Hospital Kralovske Vinohrady (EK-VP/24/O/2022) and by the respective local Ethics Committees. The trial was registered at ClinicalTrials.gov (NCT05717725). An academic steering committee designed the study, and data were collected and managed by an independent team of statisticians. The first and last authors had full access to all the data and analyses and wrote all drafts of this manuscript, with review and edits by the other authors. The authors attest to the accuracy and completeness of the data and to the fidelity of the trial conduct to the protocol. The partial funder of the study (Biotronik SE & Co. KG, which provided the ICMs) had no role in the study design, data collection, data analysis, data interpretation, or writing of reports.

Participants

The study enrolled patients >18 years of age with highly symptomatic paroxysmal or persistent AF, confirmed by 12-lead electrocardiographic or Holter monitoring, and with at least 1 symptomatic episode of AF within 6 months before randomization. In accordance with current guidelines for AF management, treatment with antiarrhythmic drugs or drug failure was not mandatory for enrolment. However, the presence of significant AF-related symptoms was ensured by requiring a significantly reduced score (≤ 50) on the Atrial Fibrillation Effect on Quality-of Life (AFEQT) questionnaire during screening.¹⁴ This 20-item questionnaire has been validated to correlate with the severity of AF-related symptoms.¹⁵ It provides a 4-item Symptoms score, an 8-item Daily Activities score, a 6-item Treatment Concerns score, and a 2-item Treatment Satisfaction scale. Overall and subscale scores range from 0 to 100, with a score of 0 indicating complete disability and 100 representing optimal quality of life.

The main exclusion criteria included structural heart disease, significant valvular disease, long-standing persistent AF, a history of tachycardia-induced cardiomyopathy, prior left atrial ablation, or other documented but untreated arrhythmias such as atrial flutter. A complete list of all inclusion and exclusion criteria is given in the [Supplemental Material](#). Participants were recruited from the outpatient departments of the participating centers. Within 7 days of providing written informed consent, participants received an ICM (Biomonitor IIM, Biotronik SE & Co KG, Germany) with remote monitoring capability.

Electrophysiology Study

All patients underwent an electrophysiology study at 30 $\pm$ 7 days after implantation of an ICM. The purpose of the electrophysiology study was to rule out unusual causes of AF such as a concealed atrioventricular bypass tract or atrioventricular nodal reentry tachycardia and to permit catheter ablation of this substrate if present. Class Ic antiarrhythmic drugs or sotalol was stopped 2 days before the procedure. Amiodarone was permitted in patients with persistent AF as a third-line antiarrhythmic drug and could be continued through the procedure. All subjects received non-vitamin K anticoagulants for at least 3 weeks before the procedure regardless of baseline stroke risk (as determined by the CHA₂DS₂-VASc score). Patients

with AF at the beginning of the electrophysiology study were cardioverted, and an electrophysiology study was performed.

The electrophysiology study was performed using 2 electrophysiology catheters inserted through the femoral vein (right side preferred). Femoral access was obtained with ultrasound guidance. One catheter was inserted in the coronary sinus, the other was inserted at the His bundle area (to record atrial, His, and ventricular activity). SVT and concealed accessory pathway were excluded with standard electrophysiology protocols: retrograde ventricular pacing and decrement and programmed atrial stimulation. If a concealed accessory pathway or inducible sustained atrioventricular reentrant tachycardia was present, the arrhythmia was targeted, and randomization was not permitted. Isolated echo beats without inducible sustained atrioventricular reentrant tachycardia were not considered as an exclusion criterion.

Randomization and Masking

Immediately after a negative electrophysiology study, patients were randomized in a 1:1 ratio to receive either PFA (PFA group; ablation) or a sham procedure (sham group; control). Randomization was performed with centralized software in permuted blocks of 4 patients, stratified for sex and AF type (paroxysmal or persistent). Patients remained blinded to their treatment assignment until study completion or early withdrawal. Blinding during the procedure was achieved through adequate deep sedation under the supervision of an anesthesiologist or as per the standard of care at each center. Sedation was maintained throughout the ablation procedure for patients in the PFA group and for 20 to 30 minutes for the sham group until sheath removal.

Operating physicians, including electrophysiologists and anesthesiologists, were not blinded to treatment assignment. However, clinical coordinators, biomedical engineers, study nurses, and physicians responsible for patient follow-up during inpatient visits and telephone checks remained blinded to group allocation. Technicians from the core laboratory analyzing ICM data were also blinded. The Bang and James Blinding Indices were used to evaluate the effectiveness of subject blinding before discharge after the procedure and again at 6 months.¹⁶

Ablation Procedure (Ablation Arm)

In patients allocated to the PFA arm, ablation was performed immediately after the electrophysiology study. The same venous access used in the electrophysiology study was used for catheter insertion. An intracardiac ultrasound catheter was positioned in the right atrium, and the transseptal puncture was carried out under intracardiac ultrasound guidance. A bolus of heparin according to body weight was delivered before transseptal puncture. Pulsed field energy applications could be started when activated clotting times were >300 seconds, and the activated clotting time was maintained >300 seconds throughout the procedure. At investigator discretion, atropine (0.5–1 mg) was given to avoid a vagal reaction. The ablation was performed with the pentaspline PFA catheter (Farawave; Boston Scientific, Inc). PFA energy was applied in standard fashion: 4 times in the basket and 4 times in the flower configuration for each pulmonary vein plus additional lesions as needed.¹⁷ PV isolation was confirmed by the presence of

entrance block using the pentaspline PFA catheter. In the event of uncertainty, at operator discretion, the completeness of PV isolation could be verified by entrance or exit block using a conventional multielectrode circular or multispline mapping catheter. In patients with persistent AF, additional posterior wall ablation could be performed with the catheter in the flower configuration.¹⁸ At the end of the procedure, a single figure-of-eight suture was placed to achieve hemostasis.

Sham Procedure (Sham, Control Arm)

Patients in the control group received ≈20 to 30 minutes of anesthesia—propofol, opioids, and benzodiazepines—administered in a manner consistent with that used in the PFA arm. At the conclusion of the procedure, figure-of-eight sutures were placed to achieve hemostasis.

Sedation During the Electrophysiology, Ablation, and Sham Procedures

Sedation during the electrophysiology study was performed similarly in both patient groups, aiming for a sedation level of −1 on the Richmond Agitation-Sedation Scale. The choice of sedative technique was left to the discretion of the attending anesthesiologist or cardiologist, as per center standard of care. Sedation was typically achieved with the infusion of propofol, along with doses of opioids (eg, sufentanil 5–10 µg) and benzodiazepines (midazolam or remimazolam) as needed as per center standard of care.

On completion of the electrophysiology study, the depth of sedation was increased. In the sham group, the target level was a Richmond Agitation-Sedation Scale score of −4, achieved by titrating higher doses of the ongoing sedative regimen. In the PFA group, sedation was further deepened to a Richmond Agitation-Sedation Scale score of −5. All procedures were performed with spontaneous ventilation and the use of oxygen masks and with the guidance of an anesthesiologist or as per the center standard of care.

Postprocedural Care

Patients were discharged the day after the procedure. Postprocedural care, including monitoring, suture removal, and anticoagulation management at discharge, was the same in both groups. All subjects were maintained on anticoagulation for at least 2 months, with continuation thereafter based on individual baseline stroke risk, as determined by the CHA₂DS₂-VASc score. The use of antiarrhythmic drugs at discharge was left to physician discretion; however, all antiarrhythmic drugs, including amiodarone, were required to be discontinued before or at least at the 2-month follow-up visit.

Follow-Up

Remote ECG monitoring was performed with the ICMs in all patients. ICM settings were adjusted after implantation to record atrial arrhythmia episodes exceeding 30 seconds, including AF, atrial tachycardia (AT), or atrial flutter (AFL). All ECGs were assessed by the core laboratory, which was blinded to treatment assignment. AF/AT/AFL burden was assessed during the index procedure (baseline AF/AT/AFL burden) and at 2, 3, and 6 months.

In-Person Visits

Patients attended in-person follow-up visits at 2, 3, and 6 months after the procedure. During each visit, electrocardiographic recordings, adverse events, clinical arrhythmia recurrences, and any hospital admissions or emergency department visits were assessed. The AFEQT questionnaire was completed at each visit. In addition, the Hospital Anxiety and Depression Scale (HADS), an assessment of psychological distress, and the blinding index questionnaire were administered at the 6-month visit.

Phone Visits

Telephone follow-ups were conducted at 1, 4, and 5 months after the procedure to assess for symptoms using the AFEQT questionnaire.

Early Cross-Over Criteria

Early cross-over (within 6 months after randomization) was permitted for patients with documented AF/AT/AFL recurrence accompanied by persistent or worsening symptoms as defined by an AFEQT score equal to or less than the baseline/index score for 3 consecutive months after randomization, a reduction in the AFEQT score by at least 10 points after the 2-month visit, or both.

Outcomes

Two co-primary outcomes were assessed at 6 months: (1) freedom from recurrent AF/AT/AFL episodes (after the 2-month blanking), assessed as time to first documented recurrence, and (2) quality of life according to the AFEQT score at 6 months after ablation or at the time of cross-over for early exits. An episode of AF/AT/AFL was defined as either (1) a recording containing at least 30 seconds of continuous interpretable signal or (2) a 12-lead ECG showing the arrhythmia throughout the entire tracing, with at least 10 seconds of continuous interpretable signal. For quality-of-life comparison, we calculated the change in AFEQT score for each patient from randomization to 6 months (or to cross-over in cases of early exit; $\Delta \text{AFEQT} = \text{AFEQT}_{6\text{months}} - \text{AFEQT}_{\text{randomization}}$) and compared these changes between the 2 groups.

Secondary end points included (1) 6-month differences in AF/AT/AFL burden, (2) 6-month differences in the HADS score, (3) correlation of AFEQT scores with AF recurrence/burden data, (4) number of patients exiting the assigned treatment, (5) unplanned cardiovascular hospitalizations, and (6) procedure-related major complications.

AF Burden and Psychological Distress Assessment

AF burden during the postblanking period (between the end of the 2-month blanking period and the 6-month visit) was compared between groups using absolute values recorded at the 6-month visit. Individual changes from baseline were not used in this comparison.

HADS is a clinically validated tool used to diagnose psychological distress and to track improvements in response to treatment.¹⁹ The tool consists of 14 questions, with 7 assessing anxiety and 7 assessing depression. Responses are answered on a 4-point scale ranging from 0 to 3, and an overall combined score, depression plus anxiety, indicates overall psychological distress (range, 0–42). Differences in HADS scores between the randomization visit (baseline) and the 6-month visit (or at

the time of cross-over for early exits) were compared between the 2 groups.

Adverse events

The study used a Clinical End-Point Committee for the review and adjudication of adverse events. These were defined as any unfavorable or unintended disease, sign, or symptom (including abnormal laboratory findings) that could be temporally associated with the procedure. The list of anticipated adverse events or serious adverse events is provided in the protocol, available in [Supplemental Material](#).

Sample Size Considerations

To ensure enrollment of the optimal number of patients from an ethical standpoint, a bayesian framework was used. This approach permits early termination of the study if clear superiority is demonstrated according to prespecified stopping rules or (a planned increase in sample size beyond the initially targeted 60 patients if superiority is not achieved at that point but remains highly probable with the inclusion of an additional 10 to 20 participants. A detailed justification for the bayesian design is provided in the study protocol. The sample size was determined adaptively in increments of 30, 40, 50, and 60 and ultimately the potential for up to 80 patients. As per the detailed description of the methodological framework,²⁰ at each enrollment milestone, the predictive probability of ultimately demonstrating superiority of the first binary co-primary end point was calculated. Although the analyses of both co-primary end points were conducted with a bayesian framework, the initial sample size estimate was derived with a traditional frequentist power analysis. Note that this approach does not contradict bayesian principles but was used to ensure that the required sample size was both achievable and operationally feasible.

For the first co-primary end point, we assumed 65% freedom from AF/AT/AFL at 6 months with PFA and a maximum of 25% freedom from AF/AT/AFL with the sham control. Accordingly, with 80% power and a significance level of $\alpha=0.025$, a total of 25 patients per group was required. For the second co-primary end point, assuming a mean $\pm$ SD AFEQT score of 50 $\pm$ 20 in the sham control and an expected improvement to a score of 70 with PFA, 25 patients per group, or 50 patients total, were required. To account for potential crossover outside the protocol, a sample size of 60 patients was required.

Statistical Analysis

The study was conducted with bayesian statistical methods on an intention-to-treat principle. Noninformative and default objective priors were used for the co-primary end-point models underlying the adaptive monitoring decisions (beta [1, 1] and default objective Jeffreys-Zellner-Siow); weakly informative priors centered at the null were used for regression and survival models to ensure numerical stability. Complete prior specifications for all parameters across all models are provided in the [Appendix in the Supplemental Material \(Table S1\)](#).

First Co-Primary End Point

The first co-primary end point was evaluated with 2 complementary Bayesian approaches. We modeled 6-month freedom from AF/AT/AFL as a binary end point using a binomial

likelihood with a conjugate beta (1, 1) prior, yielding posterior distributions for the group-specific freedom probabilities in the PFA and sham groups. These posterior distributions were used to compute posterior probabilities of superiority and interim predictive probabilities for adaptive monitoring. We modeled time to first AF/AT/AFL recurrence after the 2-month blanking period using a Cox partial likelihood with a weakly informative normal (0, 2) prior on the log-hazard ratio, yielding a posterior distribution for the treatment-specific hazard ratio. Posterior inference was obtained with hamiltonian Monte Carlo implemented in Stan (4 chains, 4000 iterations per chain, including 1000 warm-up iterations). The posterior median hazard ratio and 95% credible interval (CrI) are reported. The proportional hazards assumption was assessed with Schoenfeld residuals. Kaplan-Meier curves are presented for descriptive visualization only.

Second Co-Primary End Point

The AFEQT end point was prespecified as the individual change from randomization to 6 months (or at cross-over for early exits). We modeled the between-group differences in AFEQT change scores using a gaussian likelihood with a Jeffreys-Zellner-Siow prior (Cauchy [0, 0.707]) on the standardized effect size, yielding a posterior distribution for the mean difference and a Bayes factor, implemented in JASP software. Because a shift in AFEQT values was observed between screening and randomization, sensitivity analyses were performed defining change relative to screening and using a baseline-adjusted bayesian ANCOVA model.

Secondary End-Point Analyses

We modeled the between-group differences in HADS scores and AF burden—outcomes with clearly nonnormal distributions—using a rank-based likelihood with a Jeffreys-Zellner-Siow prior on the standardized effect size, yielding posterior distributions for the between-group differences, implemented as a bayesian Mann-Whitney *U* test in JASP. We modeled the longitudinal association between AFEQT score and AF burden using a gaussian likelihood with weakly informative priors on the fixed effects (Normal [0, 0.5] on the AF burden slope; normal [60, 20] on the intercept) and a random intercept for participant, yielding a posterior distribution for the within-individual AF burden–AFEQT slope, fitted with brms in R software. Analytical conjugate models (beta-binomial and bayesian *t* test) were computed without Markov chain Monte Carlo sampling, whereas regression and survival models were estimated with hamiltonian Monte Carlo. All statistical analyses were performed with Python 3.11, JASP 0.95.3, and R 4.5.2. Across all models, noninformative or weakly informative priors centered at no effect were used; prior classification and complete prior specifications for all parameters are provided in the [Appendix in the Supplemental Material](#). In addition, prior sensitivity analyses were performed for all primary bayesian models.

Posterior probability of superiority represents the probability, given the observed data and prespecified model assumptions, that PFA is more effective than sham.

Formal Rules for Premature Study Termination

At each enrollment milestone, after completion of the 6-month follow-up in 30, 40, and 50 patients, the predictive probabilities

of ultimately demonstrating superiority for both co-primary end points were calculated. If the probability of the first co-primary end point, freedom from AF/AT/AFL, exceeded 0.99, then the second co-primary end point, changes in AFEQT score, was computed with a bayesian 2-sample *t* test with a noninformative prior distribution. If the second co-primary end point also showed an extremely high probability of success (Bayes factor >100), the study could be terminated early (ie, no further enrollment). Conversely, if the predictive probability for eventual success fell below 0.05 for either end point at any interim analysis milestone, the trial could be stopped for futility.

If superiority was not demonstrated in either co-primary end point after analysis of 60 patients with complete 6-month follow-up, continued enrollment was permitted under the following conditions: (1) The predictive probability of success with an additional 20 patients exceeded 0.80, and (2) there was a clear and consistently increasing trend in the probability of reaching significance for the co-primary end points across prior interim analyses. The final decision to continue or terminate enrollment was the responsibility of the Data and Safety Monitoring Board (DSMB), which was established at the beginning of the study. The DSMB received not only the results of all interim analyses but also all reports on adverse events and serious adverse events.

Data Collection

All data were centrally stored on a secure web server and monitored by study information technology specialists, who checked each variable for entry, validity, and plausibility and contacted the staff in cases of incomplete data. Data on AF recurrences obtained from ICMs and analyzed by the core laboratory were not available to the personnel conducting outpatient follow-ups, thereby preserving blinding integrity. An independent team of statisticians performed interim analyses according to the pre-specified statistical analysis plan and communicated results to the steering committee and DSMB. No imputation of missing data was planned. The data underlying the study are available from the corresponding author on reasonable request.

RESULTS

Patients

From September 14, 2023, to April 9, 2025, 131 patients were screened for participation in the study (Figure 1). Ultimately, 61 patients provided informed consent, received an ICM, and completed an electrophysiology study. During the study, sustained tachycardia (atrioventricular reentrant tachycardia) was induced in 1 patient. Per the protocol, this arrhythmia was targeted for ablation, and the patient was excluded and not randomized. Thus, 60 patients were randomized to either PFA or sham. The median time from enrollment to randomization was 43 days (interquartile range, 33–46 days).

The baseline characteristics of the 2 randomized groups appeared to be balanced (Table 1). The cohort was relatively evenly split between sexes; the majority of patients (77%) had paroxysmal AF; most patients (70%) were taking antiarrhythmic drugs; and the population was a highly symptomatic one.

Procedure and Follow-Up

All patients underwent the allocated procedure. In the PFA arm, all patients underwent successful PVI, and 10 patients underwent posterior wall isolation, including all 7 patients with persistent AF and 3 with paroxysmal AF who presented for the procedure in AF. Procedural characteristics are summarized in Table 2. Two patients with concomitant perimitral flutter underwent ablation of the lateral mitral isthmus. The overall procedure time, including the baseline electrophysiology study, was well balanced between PFA and sham (79 ± 18 minutes versus 72 ± 23 minutes; $P=0.09$; Table 2). There were no procedure-related complications in either group.

The median follow-up from randomization to the 6-month clinical visit was 184 days (interquartile range, 178–187 days) and 184 days (interquartile range, 176–193 days) in the PFA and sham groups, respectively. During follow-up, 5 patients in the sham group crossed over to PFA because of arrhythmia recurrences; characteristics of the cross-over patients are shown in Table S2. There was 1 adverse event: 1 patient in the sham group was hospitalized for worsening AF and crossed over to PFA during this hospitalization. Another sham patient required a pacemaker for symptomatic bradycardia during follow-up. There were no deaths or other adverse events during follow-up. In the sham group, antiarrhythmic drugs were initiated in 4 patients (13.3%) and 6 patients (20%) at the 3-month and 6-month visits, respectively, all due to AF recurrences. No patient in the PFA group received antiarrhythmic drugs during follow-up.

Compliance with follow-up was good, with 100% of patients presenting for the clinical visits. Because all patients had received ICMs throughout the study, there were no loss to rhythm monitoring follow-up. No patients were lost to follow-up.

Interim Analyses

In the 40-patient interim analysis, the predictive probability for the first co-primary end point met the predefined threshold for early stopping (Table S3). However, the Bayes Factor for the second co-primary end point was not met. Conversely, the 50-patient interim analysis revealed that the stopping rules for both co-primary end points were met. At that point, 60 patients had already been randomized, so the DSMB recommended completing follow-up for all randomized patients but with no further enrollment.

Primary Outcomes

The first co-primary outcome, freedom from AF/AT/AFL, was achieved in 28 patients (93.3%) in the PFA group and 5 patients (16.7%) in the sham group (posterior hazard ratio, 19.6 [95% CrI, 6.7–76.9]; posterior probability of superiority >0.99; Table 3, Figures 2, and Figure S1).

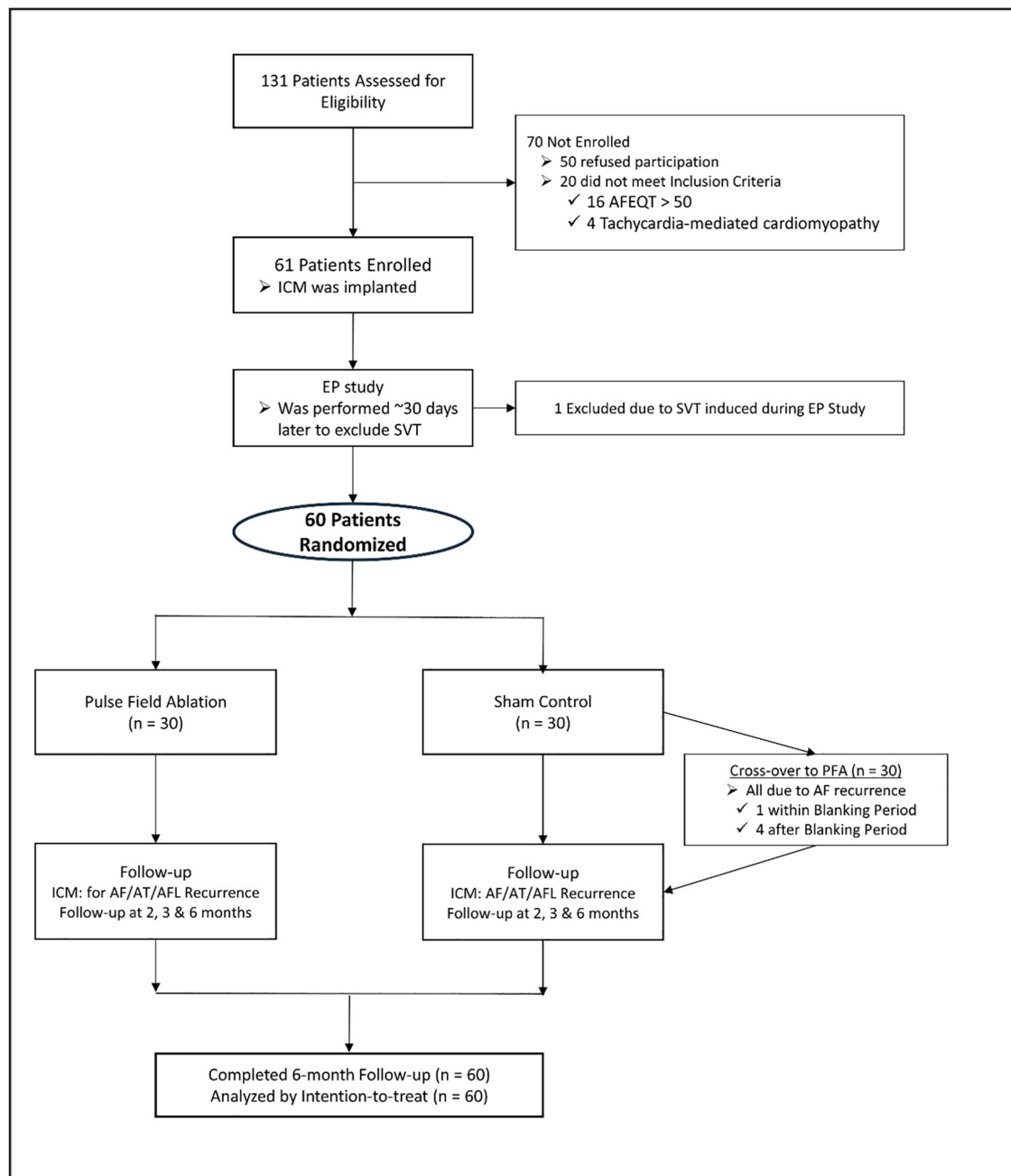


Figure 1. CONSORT Diagram.

Of the 131 patients screened for participation in the trial, 70 were classified as having screening failure: 50 patients (71%) declined to participate; 16 patients (23%) were deemed not to be symptomatic enough (Atrial Fibrillation Effect on Quality-of-Life [AFEQT] score >50); and 4 patients (6%) had tachycardia-induced cardiomyopathy. (It was felt to be unethical to include the last category of patients in this sham-controlled trial.) One patient was excluded because a supraventricular tachycardia (SVT) was induced during the electrophysiology (EP) study. Of the remaining 60 patients, making up the intention-to-treat population, 30 patients each were assigned randomly to either pulsed field ablation (PFA) or the sham control. All patients receiving ablation underwent the assigned PFA procedure, but 5 patients in the sham group crossed over to ablation because of arrhythmia recurrences. All patients completed the 6-month follow-up period. AF indicates atrial fibrillation; AFL, atrial flutter; AT, atrial tachycardia; CONSORT, Consolidated Standards of Reporting Trials; and ICM, implantable cardiac monitor.

No violation of the proportional hazards assumption was observed (global Schoenfeld test, $P=0.62$). The Kaplan-Meier curves shown in Figure 2B represent conventional nonparametric estimates for visualization; the hazard ratio and 95% CIs were derived from the bayesian Cox proportional hazards model.

The second co-primary outcome, the improvement in AFEQT score from baseline, was greater with PFA (43.9 ± 18.1) than sham (11.3 ± 27.9 ; posterior mean difference, 32.6 ± 6.3 [95% CrI, 20.2–44.9]; posterior probability of superiority >0.99; Bayes factor >100; Table 3, Figure 3, and Figures S2 and S3. In addition to these outcomes for

Table 1. Patient Characteristics at Baseline*

	PFA (n=30)	Sham procedure (n=30)	P value†
Age, y	62±10	66±9	0.093 ^W
Sex, n (%)			0.6 ^{χ²}
Male	15 (50)	13 (43)	
Female	15 (50)	17 (57)	
Body mass index, kg/m ²	29.9±6.1	29.9±4.9	0.8 ^W
CHA ₂ DS ₂ -VASc score	2.3±1.6	2.5±1.4	0.3 ^W
AF type, n (%)			>0.9 ^{χ²}
Paroxysmal	23 (77)	23 (77)	
Persistent	7 (23)	7 (23)	
Concomitant clinical conditions, n (%)			
Heart failure‡	4 (13.3)	6 (20.0)	0.5 ^{χ²}
NYHA class II	4 (100)	6 (100)	
Previous stroke or TIA	2 (6.7)	0 (0)	0.5 ^F
Coronary artery disease	2 (6.7)	0 (0)	0.5 ^F
Hypertension	23 (76.7)	24 (80.0)	0.8 ^{χ²}
Diabetes	3 (10)	7 (23.3)	0.2 ^F
Dyslipidemia	17 (56.7)	13 (43.3)	0.3 ^{χ²}
Thyroid dysfunction	5 (16.7)	4 (13.3)	>0.9 ^F
History of cardioversion	8 (26.7)	9 (30.0)	0.8 ^{χ²}
Baseline medications, n (%)			
Antiarrhythmic drugs	24 (80)	18 (60)	0.09 ^{χ²}
Propafenone	14 (47)	9 (30)	0.2 ^{χ²}
Amiodarone	9 (30)	8 (27)	>0.9 ^F
Sotalol	1 (3)	0 (0)	>0.9 ^F
Dronedarone	0 (0)	1 (3)	>0.9 ^F
Oral anticoagulants (NOAC)	30 (100)	30 (100)	—
β-Blockers	19 (63)	20 (67)	0.8 ^{χ²}
No. of antihypertension agents	1.7±1.12	1.7±1.0	>0.9 ^W
Statins	13 (43)	17 (57)	0.3 ^{χ²}
Rhythm at baseline examination, n (%)			
Sinus rhythm	24 (80)	25 (83)	0.7 ^{χ²}
AF	6 (20)	5 (17)	
Echocardiography			
Left atrial size, mm	41.3±5.6	40.4±4.7	0.5 ^W
Left atrium volume index, mL/m ²	46±21	42±15	0.3 ^W
Left ventricular ejection fraction, %	60.9±4.6	61.3±3.7	0.8 ^W
AF severity (EHRA class), n (%)			0.4 ^{χ²}
EHRA I	1 (3.3)	0 (0)	
EHRA II	18 (60.0)	15 (50.0)	
EHRA III	11 (36.7)	15 (50.0)	
AFEQT score			
At time of screening	38.1±12.7	38.8±9.5	0.7 ^W
At time of randomization	41.6±16.4	49.0±17.5	0.084 ^W
HADS score (at randomization)	10.0 (7.0–16.25)	9.5 (6.0–16.0)	0.9 ^W

AF indicates atrial fibrillation; AFEQT, Atrial Fibrillation Effect on Quality-of-Life; EHRA, European Heart Rhythm Association; HADS, Hospital Anxiety and Depression Scale; NOAC, non-vitamin K antagonist oral anticoagulant; NYHA, New York Heart Association; PFA, pulsed field ablation; and TIA, transient ischemic attack.

*Plus-minus values are mean±SD.

†P values were determined by Wilcoxon rank-sum test (W), Pearson χ^2 test (χ^2), or Fisher exact test (F).

‡Refers to heart failure with preserved ejection fraction.

Table 2. Procedural Characteristics

	PFA (n=30)	Sham (n=30)	P value
Hospital stay, mean±SD, d	1.6±0.5	1.6±0.5	0.8
Procedure duration, mean±SD, min	79±18	72±23	0.09
Anesthesia, n (%)			
General anesthesia (orotracheal intubation)	0 (0)	0 (0)	1.0
Deep sedation	30 (100)	30 (100)	
Rhythm at the beginning of the procedure, n (%)			
Sinus rhythm	24 (80)	26 (86.7)	0.7
Atrial fibrillation	6 (20)	4 (13.3)	
Electric cardioversion performed, n (%)	5 (16.7)	4 (14.8)	
Rhythm at the end of the procedure, n (%)			
Sinus rhythm	30 (100)	30 (100)	1.0
AF	0 (0)	0 (0)	
PV isolation, n (%)	30 (100)	NA	
Entrance conduction block (all PVs)	30 (100)	NA	
Left atrial posterior wall ablation, n (%)	10 (33.3)	NA	
Mitral isthmus ablation, n (%)	2 (6.7)	NA	
Cavotricuspid isthmus ablation, n (%)	0 (0)	NA	
No. of PF applications			
For PV isolation	43±9	NA	
For outside PVs	21±12	NA	

AF indicates atrial fibrillation; NA, not applicable; and PV, pulmonary vein.

the global AFEQT score, the AFEQT subdomains demonstrated similarly greater improvements with PFA than sham (Table 4 and Figure S4). The results of both co-primary

outcomes were similar also in the per-protocol analysis (Table S4 and Figure S5). Prior sensitivity analyses for all primary bayesian models are reported in the Appendix in the Supplemental Material (Table S5); all conclusions were robust to prior specification. The improvement in AFEQT score was also significantly greater in sensitivity analyses, whether assessed between screening and 6 months or at 6 months after adjustment for baseline AFEQT score measured at the time of randomization with a bayesian ANCOVA model (Table S6 and Figure S6).

Secondary Outcomes

The AF/AT/AFL burden at 6 months was significantly lower in the PFA group (median burden, 0 [0–0]) than the sham group (0.43 [0.04–3.47]; between-group median difference, −0.39% [95% CrI, −2.5 to −0.1], posterior probability of superiority >0.99; Table 3 and Figure 4). Differences in the AF/AT/AFL burdens between groups were alternatively calculated by the change in burden between 6 months and baseline. This again revealed significantly greater reductions with PFA than sham, translating to a >99.9% relative reduction and a 7.5% relative increase in AF/AT/AFL burden for the PFA and sham groups, respectively (Figure 4).

The differences in the HADS scores are shown in Table 3 and Figure 5. The HADS scores decreased by −4 points (−7.8 to −2.0) with PFA and by −0.5 (−4.5 to 1.0) in the sham group (between-group median difference, −3.5 [95% CrI, −6.0 to −1.0]; posterior probability of superiority >0.99). Last, there is a significant inverse relationship between AF recurrence/burden and

Table 3. Summary of the Results of the Primary and Key Secondary Outcomes

	PFA (n=30)	SHAM (n=30)	Between-group hazard ratio or difference	95% CrI	Posterior probability of superiority
Primary end points					
First co-primary end point, AF/AT/AFL freedom, n (%)	28 (93.3)	5 (16.7)	19.6*	6.7, 76.9	>0.9999
Second co-primary end point, AFEQT, change from baseline, mean change ± SD	43.9 ± 18.1	11.3 ± 27.9	32.6 ± 6.3†	20.2, 44.9	>0.9999 (>100‡)
Key secondary end points					
AF burden at 6 mo (%), median (IQR)	0 (0, 0)	0.43 (0.04, 3.94)	−0.39§	(−2.5, −0.1)	>0.9999
HADS, change from baseline, median (IQR)	−4 (−7.8, −2.0)	−0.5 (−4.5, 1.0)	−3.5§	(−6.0, −1.0)	0.9972
Correlation of AFEQT with AF burden	−0.25 points per 1% increase in AF burden			−0.4, −0.1	>0.9999

In a bayesian framework, the terms prior and posterior probability are used. Prior probability reflects the initial assumption about the treatment effect before observing the data. In this study, noninformative priors (beta [1, 1] for the binary model; Jeffreys-Zellner-Siow default objective prior for the *t* test) were applied for the primary monitoring decisions, implying no a priori assumption of a treatment difference. Weakly informative priors centered at no effect were used for regression and survival models—these are sufficiently diffuse to be overwhelmed by the data while ensuring numerical stability in small samples. The posterior probability represents the probability of the hypothesis being true after incorporating the observed data. In our study, it specifically represents the probability that the PFA arm is superior to the Sham procedure.

AF/AT/AFL indicates atrial fibrillation/atrial tachycardia/atrial flutter; AFEQT, Atrial Fibrillation Effect on Quality-of-Life; CrI, credible interval; HADS, Hospital Anxiety and Depression Scale; IQR, interquartile range; and PFA, pulsed field ablation.

*Posterior median hazard ratio.

†Posterior mean difference (±SD).

‡Bayes factor.

§Between-group median difference.

||Both groups, visit M2, M3, M6.

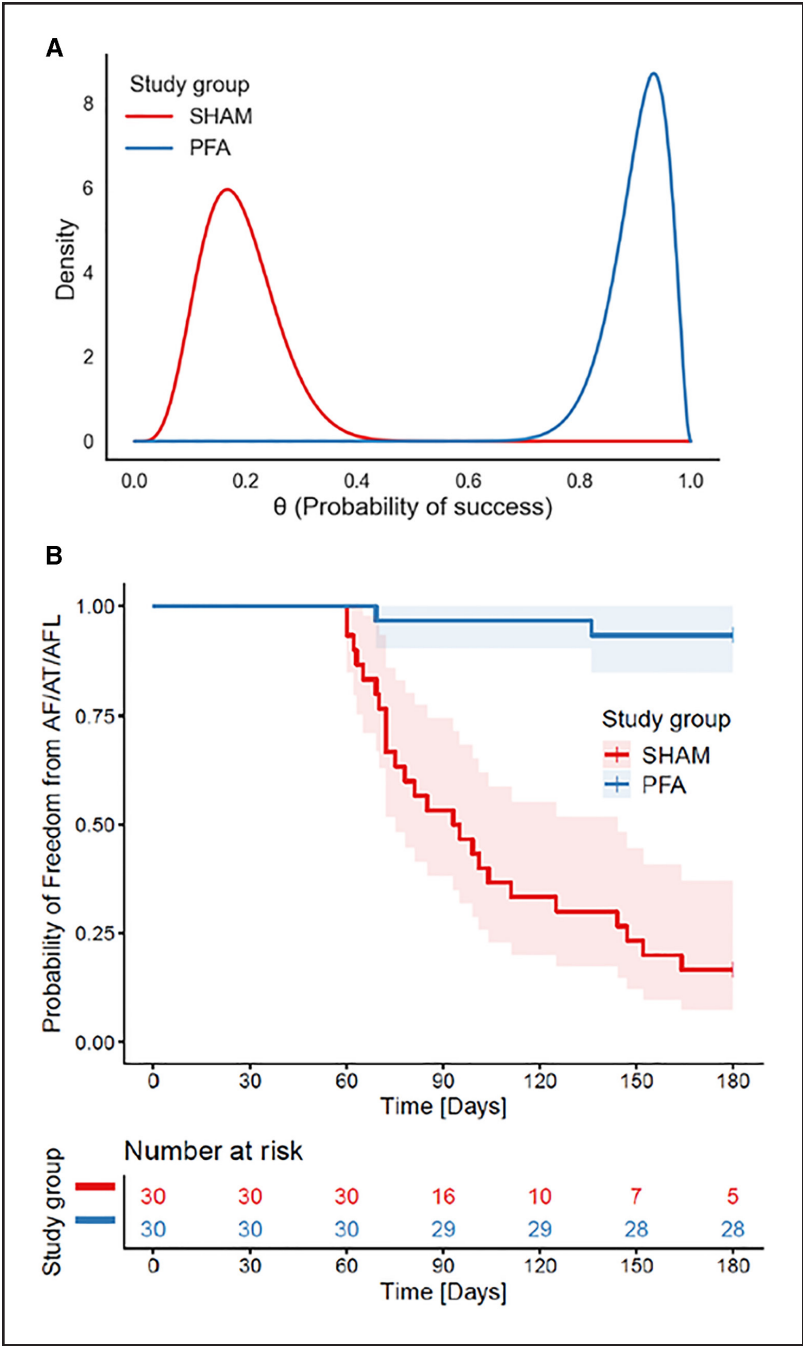


Figure 2. First co-primary end point: freedom from AF/AT/AFL.

A, Posterior distributions for the pulsed field ablation (PFA) and Sham groups. Note the complete absence of overlap between these 2 distributions. **B**, Kaplan-Meier curves of freedom from atrial fibrillation/atrial tachycardia/atrial flutter (AF/AT/AFL) after the 2-month blanking period. Shaded regions represent 95% CIs. Number at risk is shown below the x axis.

the AFEQT score, translating to a 0.25-point decrease in the AFEQT score for every 1% increase in AF burden (Table 3 and Figure S7).

With regard to the other end points, 5 patients in the sham group exited early (crossed over) from assigned treatment, of whom 1 patient underwent an unplanned cardiovascular hospitalization. There were no procedure-related complications in either group.

Blinding Assessment

The results of the Bang and James blinding indices are shown in Table S7. Taken together, they reveal that im-

mediately after the procedure, there was some degree of unblinding, but this occurred in opposite directions in the 2 groups. That is, both groups of patients, regardless of treatment assignment, believed that they received the actual treatment, a phenomenon called wishful thinking.^{21,22} Although not the same as true random guessing, the symmetrical guessing pattern of wishful thinking has the effect of canceling out these perceptions, leaving the observed treatment effect uncompromised by blinding issues.^{21,22}

Over time, there was a progressive deterioration of unblinding in the PFA group only. The combination of this asymmetric unblinding pattern and temporal evolution is

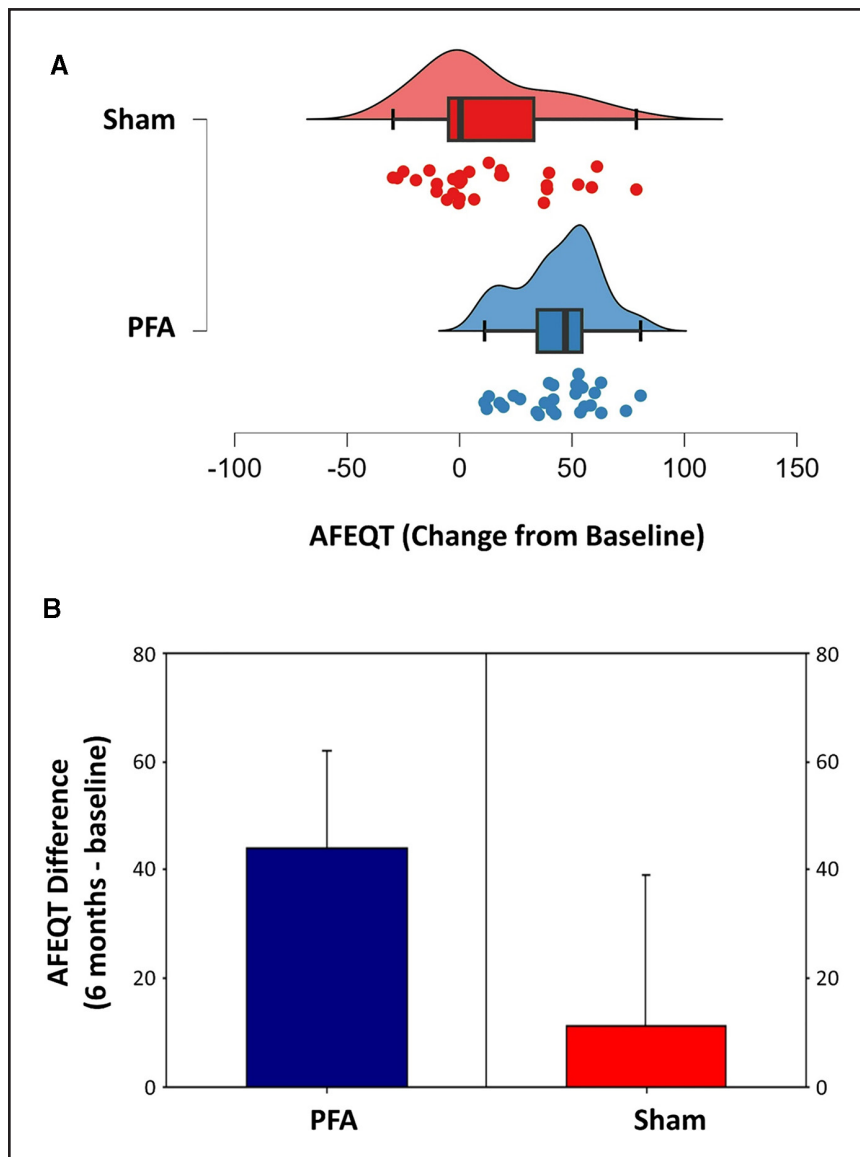


Figure 3. Second co-primary end point: changes in quality of life.

A, Posterior distributions, posterior medians with interquartile ranges, and individual data points. The difference between the pulsed field ablation (PFA) and sham groups met the criteria for superiority, with a posterior probability of >99.9%. **B**, Between-group differences (as means with SD) in individual changes of Atrial Fibrillation Effect on Quality-of-Life (AFEQT) scores between 6 month and baseline postrandomization values. This mean difference of the AFEQT scores between groups is 32.6 (95% credible interval, 20.2–44.9). AF/AT/AFL indicates atrial fibrillation/atrial tachycardia/atrial flutter.

more consistent with unblinding due to the perceivable symptomatic effects of treatment with PFA than poor study technique or flawed concealment.

DISCUSSION

In this PFA-SHAM trial, PFA to treat patients with AF showed superiority over a sham procedure for both sinus rhythm maintenance and improvement in quality of life within 6 months after the procedure. PFA was also associated with a significantly lower AF burden at 6 months as ascertained by continuous electrocardiographic monitoring with an ICM. Last, there was a marked reduction in psychological distress, as expressed by a differential improvement in the HADS score, compared between groups.

Numerous randomized trials have demonstrated the superiority of AF ablation over drugs in terms of sinus rhythm maintenance, reduced AF burden, and improved

quality of life. For instance, the EARLY-AF trial (Cryoablation or Drug Therapy for Initial Treatment of Atrial Fibrillation) and STOP-AF trial (Cryoballoon Ablation as Initial Therapy for Atrial Fibrillation), which compared cryoablation with drugs, showed an improvement in AFEQT scores by ablation by 24.4 and 35.2 points, respectively.^{3,4} In the CABANA trial (Catheter Ablation vs Antiarrhythmic Drug Therapy for Atrial Fibrillation), the improvements in quality of life, assessed with the AFEQT questionnaire, were greater after catheter ablation than with drugs, with the difference sustaining out to 5 years.²³

However, these studies compared catheter ablation with drug treatment—both active treatment strategies that may have involved placebo effects, potentially to differing extents. The placebo effects of a sham interventions have clearly been established in previous studies comparing active interventions with sham procedures.²⁴ Moreover, placebo interventions tend to exert a greater influence on patient-reported outcomes than on

Table 4. Results of Changes in the AFEQT Subdomains

	Symptoms		Daily activities		Treatment concerns	
	PFA	Sham	PFA	Sham	PFA	Sham
Randomization (mean±SD)	44.3±22.2	55.0±21.7	34.9±19.8	44±21.2	48.7±21.3	52.4±24.4
6-mo visit (mean±SD)	86.7±16.6	60.7±29.7	82±20.3	55.1±27.4	89.3±12.6	67.2±22.9
Posterior mean difference (between groups)	36.7±7.1		36.0±7.7		28.0±7.5	
Posterior probability of superiority	>0.9999		>0.9999		>0.9998	
Bayes factor for PFA superiority	>100		>100		>100	

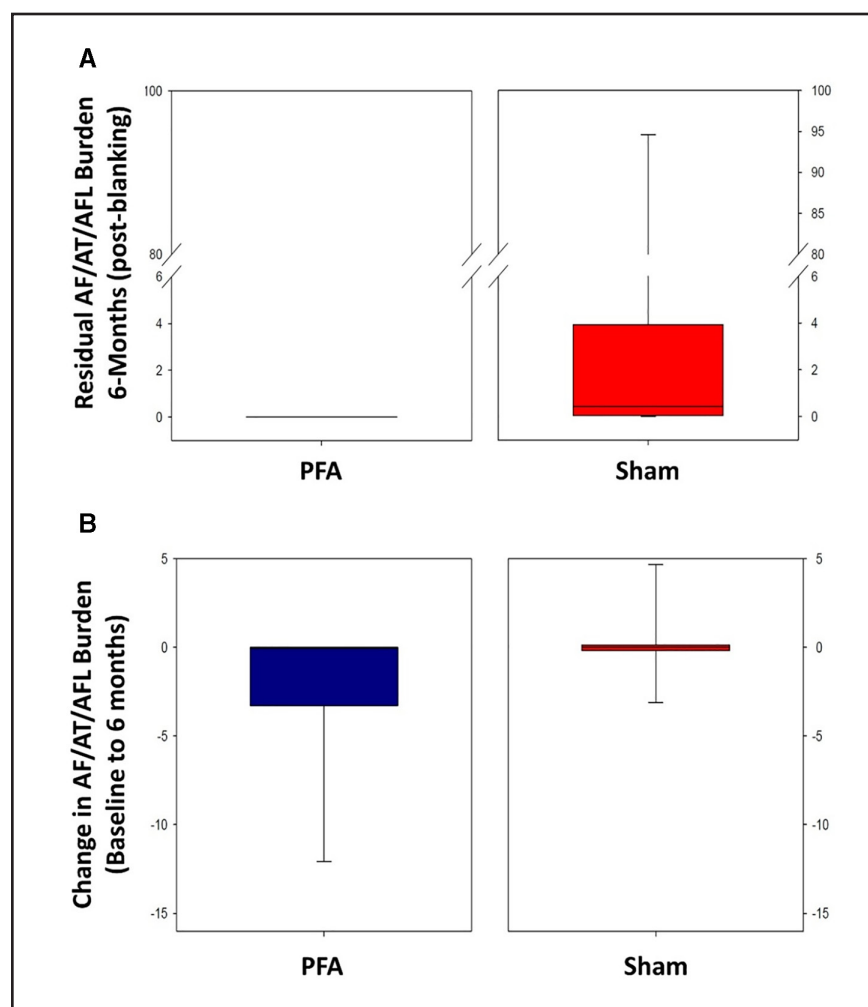
PFA indicates pulsed field ablation.

objectively measured outcomes.²⁵ Thus, in trials assessing the effect of interventional procedures, only a study comparing the intervention, in this case catheter ablation, with a sham procedure can accurately isolate and assess the actual effect of the invasive procedure.

Thus far, the only trial that compared catheter ablation using cryoballoon ablation with a sham procedure was the SHAM-PVI trial (A Randomized Sham-Controlled Study of Pulmonary Vein Isolation in Symptomatic Atrial Fibrillation).¹² The primary outcome, AF recurrence, was significantly improved with AF ablation. These results are

in complete agreement with what we observed in PFA-SHAM, albeit with our use of an arguably more contemporary ablation technology, PFA. We found that both AF recurrences as the first co-primary end point and AF burden as a secondary end point were significantly lower after PFA than with sham.

Whereas the change in quality of life was one of the secondary outcomes without prespecified power calculation in SHAM-PVI, in our study, it was a prespecified powered co-primary outcome. The quality-of-life assessment in SHAM-PVI was performed using the

**Figure 4. Change in AF/AT/AFL burden compared between groups.**

A, Absolute postprocedural residual atrial fibrillation/atrial tachycardia/atrial flutter (AF/AT/AFL) burdens between the end of the 2-month blanking period and the 6-month end point are shown. The residual burden is significantly greater for the sham group. **B**, Change in AF/AT/AFL burden calculated as the difference between the postblanking period and the preprocedural period (post-implantable cardiac monitor implantation, between screening and randomization). These absolute changes in burden (-0.04 [-3.8 to 0.0] for pulsed field ablation [PFA] and 0.005 [-0.17 to 0.15] for sham) translate to an significant relative reduction in burden by $>99.9\%$ for PFA and a nonsignificant 7.5 increase in burden for sham. Data are displayed as medians with interquartile ranges.

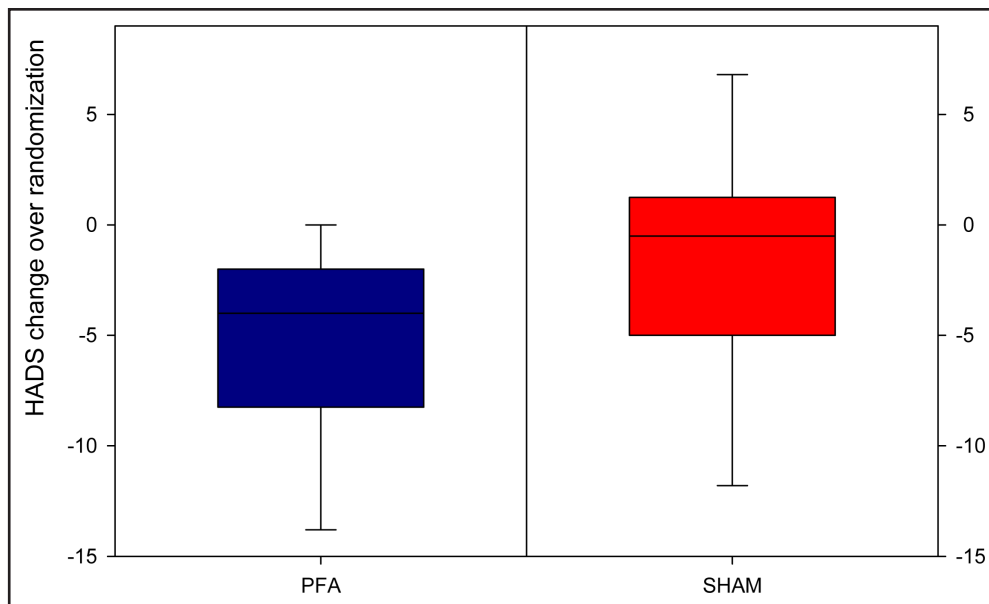


Figure 5. Change in psychological distress compared between groups.

Differences in the Hospital Anxiety and Depression Scale (HADS) scores, a measure of psychological distress, between baseline and 6 months. The reduction in the HADS score in the pulsed field ablation (PFA) group is significantly greater than the minimal change observed in the sham group. Data are displayed as median with interquartile ranges.

same AFEQT questionnaire and the same 6-month time frame as the PFA group. The AFEQT score improved by ≈ 7 points in the sham group, a change similar to that observed in our sham patients, who demonstrated an ≈ 11 -point improvement at 6 months. However, the improvement in the ablation arms of SHAM-PVI was ≈ 24 points, a value dwarfed by the ≈ 44 -point improvement we observed between randomization and 6 months. For perspective, prior studies have demonstrated that a 19-point change in the AFEQT score represents a moderate improvement in quality of life.^{26,27} Indeed, the magnitude of this effect also exceeded those reported in previous randomized trials comparing catheter ablation with drug therapy. This more pronounced improvement in our study is likely attributable to a unique aspect of PFA-SHAM: the inclusion of only highly symptomatic patients (ie, with and AFEQT score ≤ 50) to create a more efficient sham-controlled trial and to minimize the potential number of patients exposed to a sham procedure.

In our study, only 23% of patients had persistent AF, and electric cardioversion was only performed in 14.8% of the sham cohort. Accordingly, sinus rhythm maintenance in PFA-SHAM very likely accurately and directly reflects the effect of the intervention (either PFA or sham) and does not reflect adjunctive interventions such as cardioversion, a confounding issue in the SHAM-PVI trial in which 79% of the cohort had persistent AF, with cardioversion needed in 78%.

Unlike most AF ablation trials that disproportionately enroll men, our study was better balanced for sex;

women made up 53% of participants. On the other hand, the study was conducted in the Czech Republic, so based on the available demographic, the trial was exclusively a White cohort. This may be important because some data indicate differential placebo effects by race or ethnicity.²⁸

Several nonrandomized studies have suggested the positive effect of AF ablation on measures of anxiety and depression,²⁹ but the REMEDIAL trial (Randomized Evaluation of the Impact of Catheter Ablation on Psychological Distress in Atrial Fibrillation) is the only randomized trial that has compared the effect of PVI with drug treatment on psychological distress.³⁰ Using the HADS questionnaire, REMEDIAL reported that ablation was associated with significant improvement in psychological distress scores at 12 months compared with drug treatment, with a between-group difference of 4.2 points (95% CI, 7.0–1.3; $P=0.005$).³⁰ However, REMEDIAL was an open-label trial with outcomes susceptible to placebo effects. Indeed, psychological end points such as anxiety are among the most susceptible to placebo influence in clinical trials. However, in our sham-controlled study, we nonetheless observed significant improvements in the HADS score with PFA, with a between-group difference of -3.5 points (95% CrI, -6.0 to -1.0). Because the minimal clinically significant difference for HADS scores in patients with cardiovascular disease is considered to be 1.7 points,¹⁹ this decrease should be considered clinically relevant. Together, these data support a significant clinical benefit of catheter ablation in alleviating the psychological distress associated with AF.

It is important to note that our trial confirms the therapeutic efficacy of PVI using pulsed field technology. This is particularly important because PFA, by virtue of its workflow and safety benefits, has recently become the dominant energy modality being used for AF ablation in the United States, with a similar trend being observed worldwide. Because sham-controlled studies isolate the actual procedural effect, our results can also be used for planning future trials with clinical end points.

Limitations

The follow-up period in this study was limited to 6 months, which precludes a differential assessment of long-term outcomes. However, forcing highly symptomatic patients to stay in a sham group raises practical and ethical considerations. The study was conducted at only 2 centers; however, there were >5 different operators performing these procedures. By focusing our efforts and owing to the relative short follow-up, we were able to achieve complete follow-up with no patient loss and to maintain a high standard of data quality. Furthermore, given the uniformity in outcome afforded by PFA as the ablation modality, there is little reason to believe that adding more sites would affect the study outcomes. Only 60 patients were enrolled in this trial; however, the unique trial strategy of enriching for highly symptomatic patients allowed these significant results with only this modest number of patients and at the same time minimizing exposure to a sham procedure. Indeed, sensitivity analyses examining the effect of including more patients in the trial would not have improved the confidence of trial outcomes (see discussion in the [Supplemental Material](#) regarding sample size). Last, there was some degree of unblinding in the trial, which may have influenced study outcomes, particularly those of a subjective nature. However, much of this unblinding occurred over the course of the trial as the patients' treatment effects diverged. This is unsurprising because this highly symptomatic cohort experienced either an abrupt cessation of symptoms (PFA group) or continued symptoms (sham group).

CONCLUSIONS

Our trial confirms the therapeutic efficacy of PVI with PFA compared with a sham procedure in both maintaining sinus rhythm and improving patient-reported outcomes, including quality of life and anxiety.

ARTICLE INFORMATION

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Supplemental Material

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